

Elimu Predict School Mgmt System

API Integration Guide

Version 1.0 • Spring Boot Backend • JWT Secured

1. Overview

This document is the official API integration reference for the frontend development team. The backend is a Spring Boot REST API secured with JSON Web Tokens (JWT). All requests (except login and register) must include a valid Bearer token in the Authorization header.

Property	Value
Backend Framework	Spring Boot
Auth Mechanism	JWT (Bearer Token)
API Version	v1
Base URL (Local)	http://localhost:8080
Base URL (Production)	https://elimu-predict-backend.onrender.com
Content Type	application/json

2. Authentication

All protected endpoints require the JWT token to be sent in the Authorization header as shown below:

Header	Authorization: Bearer <your_token_here>
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2.1 Login

Call the login endpoint to obtain a token. The token must be stored (e.g. in memory or localStorage) and attached to every subsequent request.

```
POST /api/v1/auth/login
```

Request Body:

```
{
  "username": "your_username",
```

```
"password": "your_password"
}
```

Response:

```
{
  "token": "eyJhbGciOiJIUzI1NiJ9...",
  "username": "john.doe",
  "role": "TEACHER",
  "message": "Login successful"
}
```

2.2 Token Usage

After login, extract the token and include it in all API requests:

```
// Store the token after login
const token = response.data.token;
localStorage.setItem("jwt_token", token);
```

```
// Attach to every request (e.g. with Axios)
axios.defaults.headers.common["Authorization"] = `Bearer ${token}`;
```

2.3 Available Roles

The system supports the following user roles. Each role has access to different endpoints and dashboards:

Role	Description	Primary Dashboard
TEACHER	Class teacher — views at-risk students	/api/v1/dashboard/teacher
SENIOR_TEACHER	Senior teacher — resource planning	/api/v1/dashboard/senior
PRINCIPAL	School principal — full school overview	/api/v1/dashboard/principal
DEPUTY_PRINCIPAL	Deputy principal — school overview	/api/v1/dashboard/principal
PARENT	Parent — views their children only	/api/v1/dashboard/parent
ADMIN	Admin — full user management	/api/v1/admin/users
IT_HANDLER	IT support — system management	/api/v1/admin/users

3. Endpoint Reference

3.1 Auth Endpoints

Method	Endpoint	Description	Auth Required
POST	/api/v1/auth/login	Login and get JWT token	No
POST	/api/v1/auth/register	Register a new user	No
GET	/api/v1/auth/me	Get current user info	Yes

3.2 User Endpoints

Method	Endpoint	Description
GET	/api/v1/users	Get all users
GET	/api/v1/users/{userId}	Get user by ID
GET	/api/v1/users/role/{role}	Get users by role
POST	/api/v1/users/register	Register a user
PUT	/api/v1/users/{id}/assign-role	Assign role (query: role)
PUT	/api/v1/users/{id}/revoke	Revoke user access
PUT	/api/v1/users/{id}/restore	Restore user access
PUT	/api/v1/users/{parentId}/link-student/{admissionNumber}	Link parent to student

3.3 Student Endpoints

Method	Endpoint	Description
POST	/api/v1/students	Register a new student
GET	/api/v1/students/{admissionNumber}	Get student by admission number
PUT	/api/v1/students/{admissionNumber}	Update student details
GET	/api/v1/students/class/{className}	Get students by class
GET	/api/v1/students/parent/{parentId}	Get students linked to a parent

3.4 Marks Endpoints

Method	Endpoint	Description
POST	/api/v1/marks	Upload marks for one student

POST	/api/v1/marks/bulk	Bulk upload marks
PUT	/api/v1/marks/{id}	Update marks (query: marks)
GET	/api/v1/marks/student/{admissionNumber}	All marks for a student
GET	/api/v1/marks/student/{admissionNumber}/term/{term}	Marks by student and term
GET	/api/v1/marks/class/subject/{subjectId}/term/{term}/year/{year}	Class marks by subject/term/year

3.5 Dashboard Endpoints

Each dashboard endpoint returns data scoped to the authenticated user. Always pass term and academicYear as query parameters.

Endpoint	Role	Response Type
/api/v1/dashboard/teacher	TEACHER	TeacherDashboardDTO
/api/v1/dashboard/senior	SENIOR_TEACHER	SeniorDashboardDTO
/api/v1/dashboard/principal	PRINCIPAL / DEPUTY_PRINCIPAL	SchoolOverviewDTO
/api/v1/dashboard/parent	PARENT	ParentDashboardDTO
/api/v1/dashboard/parent/child/{admissionNumber}	PARENT	StudentReportDTO

3.6 AI Analysis Endpoints

Method	Endpoint	Description
POST	/api/v1/ai/analyze/student/{admissionNumber}	Trigger AI analysis for one student
POST	/api/v1/ai/analyze/class/{className}	Trigger AI analysis for whole class
GET	/api/v1/ai/results/student/{admissionNumber}	Get AI results by student & term
GET	/api/v1/ai/results/student/{admissionNumber}/all	Get all AI results for a student

4. Enum Reference

Use these exact string values in request bodies and query parameters:

Field	Allowed Values
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term	TERM_1, TERM_2, TERM_3
examType	CAT_1, CAT_2, CAT_3, EXAM_1, EXAM_2
role	TEACHER, SENIOR_TEACHER, ADMIN, PRINCIPAL, DEPUTY_PRINCIPAL, PARENT, IT_HANDLER

5. Sample Request Bodies

5.1 Register User

```
POST /api/v1/users/register
{
  "fullName": "Jane Smith",
  "userName": "janesmith",
  "password": "securepass123",
  "role": "TEACHER",
  "admissionNumbers": []
}
```

5.2 Register Student

```
POST /api/v1/students
{
  "admissionNumber": "ADM001",
  "fullName": "Alice Wanjiru",
  "className": "Form 1A",
  "parentId": null,
  "enrollmentYear": 2026
}
```

5.3 Upload Marks (Single)

```
POST /api/v1/marks
{
  "admissionNumber": "ADM001",
  "subjectId": 1,
  "marksObtained": 75.5,
  "examType": "CAT_1",
  "term": "TERM_1",
  "academicYear": 2026
}
```

5.4 Bulk Upload Marks

```
POST /api/v1/marks/bulk
{
  "records": [
    { "admissionNumber": "ADM001", "subjectId": 1, "marksObtained": 75.5,
      "examType": "CAT_1", "term": "TERM_1", "academicYear": 2024 },
    { "admissionNumber": "ADM002", "subjectId": 1, "marksObtained": 60.0,
      "examType": "CAT_1", "term": "TERM_1", "academicYear": 2026 }
  ]
}
```

6. Frontend Setup Checklist

Step 1 — Import the Postman Collection

1. Open Postman and click Import
2. Drag in SchoolApp.postman_collection.json
3. Update the base_url variable to match your environment
4. Run Auth → Login — the token is auto-saved to {{jwt_token}}

Step 2 — Configure CORS

The Spring Boot backend must allow requests from your frontend's origin. Ask the backend team to whitelist your development URL (e.g. <http://localhost:3000>) and your production domain.

Step 3 — Handle Token Expiry

The JWT token has an expiry time. When a request returns HTTP 401 Unauthorized, redirect the user to the login screen and clear the stored token.

Step 4 — Role-Based Routing

After login, read the role field from the auth response and route the user to the correct dashboard:

Role	Redirect To
TEACHER	/dashboard/teacher
SENIOR_TEACHER	/dashboard/senior
PRINCIPAL / DEPUTY_PRINCIPAL	/dashboard/principal
PARENT	/dashboard/parent

ADMIN / IT_HANDLER	/admin
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Test Data

Role	Username	Password	Name
ADMIN	ADMIN001	admin@123	System Admin
IT_HANDLER	ITH001	ithandler@123	IT Handler
TEACHER	TCH001	teacher123	John Lagat
TEACHER	TCH002	teacher123	Jane Nekesa
SENIOR_TEACHER	SNR001	senior123	James Okutoyi
DEPUTY_PRINCIPAL	DEP001	deputy123	Aaron Mutua
PRINCIPAL	PRI001	principal123	Lovingstone Ochieng'
PARENT	PAR001	parent123	Herine Adhiambo

7. Support & Contact

For backend issues or questions about specific endpoints, contact the backend development team or whatsapp +254796791558.